

# EC400: Solutions to Exam

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1. Denote Lagrange multiplier with  $\lambda_t$  and write

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[ \beta^t \log C_t + \lambda_t (K_t^\alpha - C_t - K_{t+1}) \right].$$

The first-order conditions wrt  $C_t$  and  $K_{t+1}$  are

$$\begin{aligned} \beta^t \frac{1}{C_t} &= \lambda_t, \\ \lambda_t &= \lambda_{t+1} \alpha K_{t+1}^{\alpha-1}. \end{aligned}$$

2. Combine together the previous equations to obtain

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} \alpha K_{t+1}^{\alpha-1}.$$

Given that  $\lambda_t$  is a shadow price of  $K_t$ , it follows that in a finite-period model, the terminal condition is  $\lambda_T K_{T+1} = 0$ . The infinite-period analog is

$$\lim_{t \rightarrow \infty} \beta^t \frac{K_{t+1}}{C_t} = 0.$$

3. Consider the following deviation from the optimal path: the entrepreneur increases consumption in period  $t$  by  $\Delta$ , which lowers the next period stock of capital by  $\Delta$ . To bring  $K_{t+2}$  back to the original path, one needs to lower  $C_{t+1}$  by  $\alpha K_{t+1}^{\alpha-1} \Delta$ . Such perturbation increases welfare in period  $t$  due to higher  $C_t$ , but decreases welfare in periods  $t+1$  because of lower  $C_{t+1}$ . The net effect evaluated using marginal utilities  $\frac{1}{C_t}$  and  $\frac{\beta}{C_{t+1}}$  is zero when the Euler condition from above is satisfied.

4. At the interior steady state,  $C_t = C_{t+1} = \bar{C} \neq 0$  and the Euler equation can be solved for  $\bar{K} = (\alpha\beta)^{\frac{1}{1-\alpha}}$ . From the resource constraint, we get  $\bar{C} = \bar{K}^\alpha - \bar{K}$ .
5. The Bellman equation is given by

$$V(K) = \max_{K'} \left\{ \log(K^\alpha - K') + \beta V(K') \right\}.$$

6. Substitute the guess into the Bellman equation

$$V(K) = \max_{K'} \left\{ \log(K^\alpha - K') + \beta(A \log K' + B) \right\}$$

and take the first-order condition

$$\frac{1}{K^\alpha - K'} = \beta A \frac{1}{K'}.$$

Therefore,

$$K' = \frac{\beta A}{1 + \beta A} K^\alpha.$$

Substitute this expression back into the Bellman equation

$$\begin{aligned} V(K) &= \log \left( K^\alpha - \frac{\beta A}{1 + \beta A} K^\alpha \right) + \beta A \log \left( \frac{\beta A}{1 + \beta A} K^\alpha \right) + \beta B \\ &= (1 + \beta A) \alpha \log K + \log \frac{1}{1 + \beta A} + \beta A \log \frac{\beta A}{1 + \beta A} + \beta B. \end{aligned}$$

It follows that

$$\begin{aligned} A &= \frac{\alpha}{1 - \alpha\beta}, \\ B &= \frac{1}{1 - \beta} \left[ \log \frac{1}{1 + \beta A} + \beta A \log \frac{\beta A}{1 + \beta A} \right] = \frac{1}{1 - \beta} \left[ \frac{\alpha\beta}{1 - \alpha\beta} \log(\alpha\beta) + \log(1 - \alpha\beta) \right] \end{aligned}$$

and the policy function is  $K'(K) = \frac{\beta A}{1 + \beta A} K^\alpha = \alpha\beta K^\alpha$ .

7. Divide both sides of the equation by  $\Delta$

$$\frac{K_{t+\Delta} - K_t}{\Delta} = K_t^\alpha - K_t - C_t$$

and take the limit  $\Delta \rightarrow 0$

$$\dot{K}_t = K_t^\alpha - K_t - C_t.$$

8. Denote a co-state with  $\lambda_t$  and write down the Hamiltonian

$$\mathcal{H}_t = e^{-\rho t} \log C_t + \lambda_t (K_t^\alpha - K_t - C_t)$$

Use the maximum principle:

$$\begin{aligned} 0 &= \frac{\partial \mathcal{H}_t}{\partial C_t} = e^{-\rho t} \frac{1}{C_t} - \lambda_t, \\ -\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial K_t} = \lambda_t (\alpha K_t^{\alpha-1} - 1), \\ 0 &= \lim_{t \rightarrow \infty} \lambda_t K_t. \end{aligned}$$

9. Substitute the first equation into the second one:

$$\rho e^{-\rho t} \frac{1}{C_t} + e^{-\rho t} \frac{\dot{C}_t}{C_t^2} = e^{-\rho t} \frac{1}{C_t} (\alpha K_t^{\alpha-1} - 1)$$

and simplify to get the Euler equation

$$\frac{\dot{C}_t}{C_t} = \alpha K_t^{\alpha-1} - 1 - \rho.$$

Intuitively, consumption is increasing in time and the entrepreneur invests more at the beginning when marginal returns on investment  $\alpha K_t^{\alpha-1} - 1$  are high and the discounting of the future  $\rho$  is low.

10. The HJB equation is given by

$$\rho V(K) = \max_C \left\{ \log C + V'(K) (K^\alpha - K - C) \right\}.$$

The first-order condition

$$\frac{1}{C_t} = V'(K_t)$$

can be differentiated wrt time to obtain

$$-\frac{\dot{C}_t}{C_t^2} = V''(K_t) \dot{K}_t.$$

Substitute these equations into the envelope condition

$$\rho V'(K_t) = V''(K_t)\dot{K}_t + V'(K_t)(\alpha K_t^{\alpha-1} - 1)$$

to substitute out the value function

$$\rho \frac{1}{C_t} = -\frac{\dot{C}_t}{C_t^2} + \frac{1}{C_t}(\alpha K_t^{\alpha-1} - 1)$$

and to get the Euler equation

$$\frac{\dot{C}_t}{C_t} = \alpha K_t^{\alpha-1} - 1 - \rho.$$